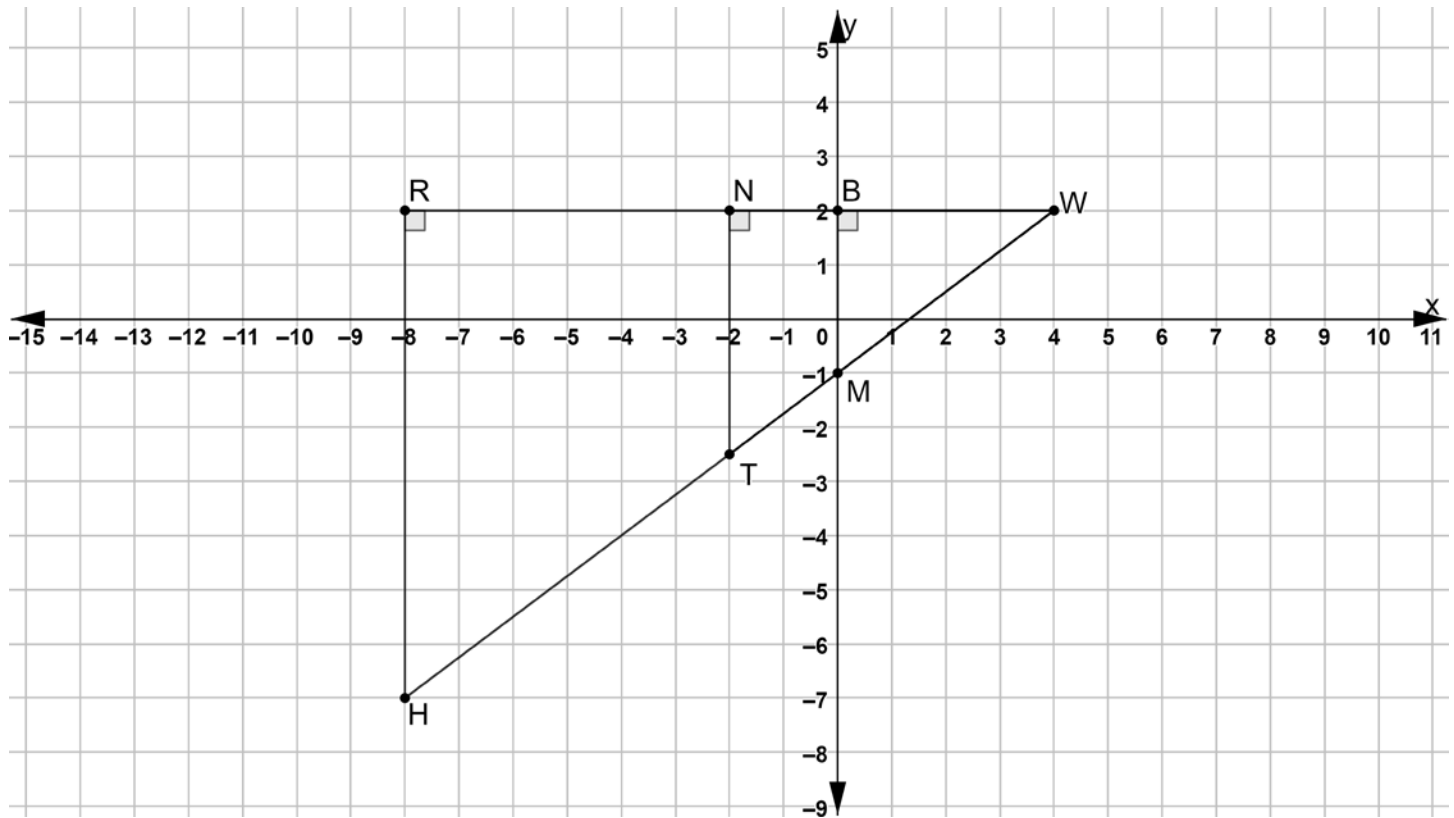


Three right triangles are shown on the coordinate grid.



The coordinates of the vertices are shown.

$R(-8, 2)$

$N(-2, 2)$

$B(0, 2)$

$W(4, 2)$

$H(-8, -7)$

$T(-2, -2.5)$

$M(0, -1)$

1. Explain why these triangles are similar.

All three triangles share $\angle W$ and have a right angle

Additionally, all sides are proportional.

Complete the table.

	Triangle	Side	Length
2.	<i>BWM</i>	$\overline{BW}$	4
		$\overline{BM}$	3
		$\overline{MW}$	5 $4^2 + 3^2 = c^2$
3.	<i>NWT</i>	$\overline{NW}$	6
		$\overline{TN}$	4.5
		$\overline{TW}$	7.5 $6^2 + 4.5^2 = c^2$
4.	<i>RWH</i>	$\overline{RW}$	12
		$\overline{RH}$	9
		$\overline{HW}$	15 $12^2 + 9^2 = c^2$

Using $\angle W$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

	Triangle	Cosine	Sine	Tangent
5.	<i>BWM</i>	$\frac{A}{H} = \frac{4}{5} = 0.8$	$\frac{O}{H} = \frac{3}{5} = 0.6$	$\frac{O}{A} = \frac{3}{4} = 0.75$
6.	<i>NWT</i>	$\frac{A}{H} = \frac{6}{7.5} = 0.8$	$\frac{O}{H} = \frac{4.5}{7.5} = 0.6$	$\frac{O}{A} = \frac{4.5}{6} = 0.75$
7.	<i>RWH</i>	$\frac{A}{H} = \frac{12}{15} = 0.8$	$\frac{O}{H} = \frac{9}{15} = 0.6$	$\frac{O}{A} = \frac{9}{12} = 0.75$

Use your calculator to find each. Round to the nearest thousandth.

	Trigonometric Ratio	Value rounded to nearest thousandth
8.	$\cos(37^\circ)$	0.799
9.	$\sin(37^\circ)$	0.602
10.	$\tan(37^\circ)$	0.754